

FLAG ISLAND, MN, USA

WMO: 727467

Lat: 49.318N

Lon: 94.903W

Elev: 1080

StdP: 14.13

Time Zone: -6.00 (NAC)

Period: 99-19

WBAN: 04926

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-23.6	-17.9	-29.0	1.1	-23.9	-25.6	1.4	-18.3	28.9	11.5	26.7	10.0	5.3	310	0.627

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.2	81.3	68.5	79.1	67.4	76.5	66.5	72.3	78.2	70.4	76.5	68.3	74.4	7.9	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
70.3	116.2	77.2	68.1	107.3	75.2	65.9	99.6	73.3	36.9	78.4	35.0	76.5	33.3	74.7	78.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
26.8	24.4	21.1	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-29.9	85.6	4.6	2.6	-33.3	87.4	-36.0	89.0	-38.6	90.4	-42.0	92.3
			-29.9	74.3	4.4	1.9	-33.1	75.7	-35.7	76.8	-38.1	77.8	-41.3	79.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	39.0	6.0	7.7	22.3	37.2	51.2	62.7	68.9	66.7	58.2	44.0	28.8	11.9	(6)
(7)		DBStd	24.35	14.14	12.61	12.70	9.48	8.44	6.32	5.23	5.48	7.69	8.36	10.32	13.04	(7)
(8)		HDD50	5951	1365	1185	858	393	88	2	0	0	18	225	637	1180	(8)
(9)		HDD65	9820	1830	1605	1323	835	431	115	24	46	228	652	1086	1645	(9)
(10)		CDD50	1921	0	0	0	7	126	383	585	518	263	39	0	0	(10)
(11)		CDD65	315	0	0	0	0	4	46	144	98	23	0	0	0	(11)
(12)		CDH74	1102	0	0	0	0	13	131	580	341	37	0	0	0	(12)
(13)		CDH80	123	0	0	0	0	1	11	69	41	1	0	0	0	(13)
(14)	Wind	WSAvg	9.5	9.4	8.9	9.1	9.1	9.9	8.5	8.4	7.9	10.1	11.7	11.6	9.6	(14)
(15)	Precipitation	PrecAvg	27.4	1.2	0.9	1.3	1.5	3.2	3.9	4.1	3.0	3.2	2.3	1.6	1.2	(15)
(16)		PrecMax	34.1	2.2	1.9	2.3	2.8	6.2	8.1	7.9	6.3	6.0	5.0	4.0	2.2	(16)
(17)		PrecMin	18.1	0.4	0.2	0.4	0.6	1.2	1.9	1.5	0.2	1.0	0.4	0.5	0.5	(17)
(18)		PrecStd	4.5	0.4	0.4	0.6	0.6	1.5	1.7	1.6	1.5	1.5	1.1	0.9	0.4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.4	39.3	54.1	63.5	75.2	81.5	84.4	84.0	78.9	66.5	53.5	37.2	(19)
(20)		MCWB	33.9	34.9	45.3	48.6	60.2	65.5	69.8	70.2	66.7	57.5	46.6	34.9	(20)	
(21)		2%	DB	34.1	36.1	45.4	57.2	71.8	78.6	81.8	80.9	73.7	63.3	47.9	34.3	(21)
(22)		MCWB	32.6	33.6	39.6	45.8	58.3	64.0	68.9	69.5	64.6	55.5	43.9	32.6	(22)	
(23)		5%	DB	29.8	32.1	42.5	54.2	66.3	74.6	79.5	78.6	72.1	58.8	45.2	32.1	(23)
(24)		MCWB	29.3	30.6	37.6	44.6	57.1	64.0	67.7	68.3	64.1	52.1	41.4	31.1	(24)	
(25)		10%	DB	26.7	27.6	37.4	50.3	63.5	72.3	77.2	74.9	69.8	55.0	42.8	29.5	(25)
(26)		MCWB	24.9	25.8	34.3	42.8	55.0	63.0	67.1	66.3	63.4	49.9	39.3	29.0	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	35.2	36.3	46.2	50.6	65.4	71.1	74.8	74.6	69.7	59.7	47.5	35.5	(27)
(28)		MCDWB	37.8	38.7	50.5	59.2	70.9	76.9	80.8	80.0	75.4	66.3	51.1	37.1	(28)	
(29)		2%	WB	32.3	33.4	41.0	48.4	61.5	67.9	72.8	72.2	67.2	56.6	44.5	32.7	(29)
(30)		MCDWB	33.7	35.4	45.6	55.7	67.3	73.7	78.8	78.1	72.3	61.9	46.9	33.8	(30)	
(31)		5%	WB	28.7	30.2	37.5	45.7	58.6	66.1	71.0	70.0	65.7	53.7	41.6	30.7	(31)
(32)		MCDWB	29.2	31.8	40.8	52.4	65.0	71.8	77.0	76.2	70.4	58.2	45.2	31.6	(32)	
(33)		10%	WB	24.6	25.9	35.1	42.7	55.8	64.2	69.0	67.9	63.5	50.3	39.2	28.2	(33)
(34)		MCDWB	26.5	27.7	38.0	49.6	62.2	69.6	75.3	73.7	67.6	54.8	42.5	28.8	(34)	
(35)	Mean Daily Temperature Range		MDBR	16.3	18.9	18.1	15.1	13.4	12.0	12.2	11.9	11.3	9.5	9.2	12.7	(35)
(36)		5% DB	MCDBR	16.7	17.8	18.5	20.1	17.9	14.6	14.0	14.2	13.9	13.4	11.1	13.4	(36)
(37)			MCWBR	15.1	15.0	13.4	11.4	9.2	8.2	8.2	8.4	8.4	8.3	7.9	12.1	(37)
(38)			MCDBR	16.5	16.7	16.8	18.5	14.8	12.9	12.2	12.6	12.0	12.4	10.5	12.7	(38)
(39)		5% WB	MCWBR	15.1	14.4	12.6	11.7	9.1	8.7	8.6	8.5	8.3	8.7	7.8	11.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.250	0.268	0.294	0.335	0.352	0.373	0.378	0.370	0.346	0.304	0.272	0.248	(40)	
(41)		taud	2.323	2.286	2.389	2.422	2.441	2.405	2.416	2.453	2.502	2.585	2.487	2.353	(41)	
(42)		Ebn at Noon	260	282	293	290	288	281	278	275	272	269	253	245	(42)	
(43)		Edh at Noon	21	28	31	33	34	36	35	32	28	22	18	18	(43)	
(44)	All-Sky Solar Radiation	RadAvg	419	791	1171	1510	1706	1785	1896	1574	1160	674	367	294	(44)	
(45)		RadStd	44	62	115	145	118	144	100	118	98	96	46	33	(45)	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (23 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page